

Bridging the Bandwidth Gap: Broadband Infrastructure Expansion and Rural Civic Participation

Prof. Sarah Jenkins & Dr. Aris Thorne | Center for Rural Policy & Digital Governance

Abstract: High-speed internet access is increasingly framed as a prerequisite for modern democratic engagement. This study evaluates the empirical effect of municipal broadband infrastructure expansion on local civic participation in rural America, tracking both online municipal portal usage and offline town hall meeting attendance.

1. Research Question

Research Question: How does the introduction of high-speed fiber broadband infrastructure in rural counties affect formal offline municipal meeting attendance and public comment submissions?

2. Empirical Methodology & Data Collection

Empirical Methodology & Data Collection: We compiled a panel dataset spanning 320 rural counties across 5 US states from 2018 to 2024. Utilizing a difference-in-differences (DiD) econometric framework, we exploited staggered county-level broadband deployments. Primary data sources include federal broadband coverage maps, municipal town hall attendance logs (N = 1,420 meetings), and public comment portal archives.

Table 1: Empirical Findings Summary

County Category	Counties (N)	Pre-Broadband Attend.	Post-Broadband Attend.	DiD Estimate
Control (Unconnected Rural)	140	14.2 per meeting	14.8 per meeting	Baseline
Treatment (Fiber Deployed)	180	13.9 per meeting	21.5 per meeting	+6.9** (p<0.01)
Digital Comment Submissions	180	2.1 per meeting	18.4 per meeting	+15.8*** (p<0.001)

3. Discussion & Implications

Discussion & Conclusion: Results demonstrate that broadband adoption substantially boosts rural civic involvement. Beyond digital submission growth (+15.8 comments/meeting), physical attendance at town hall meetings increased by 49.6% (+6.9 attendees/meeting). The findings refute the displacement hypothesis—which posits online media replaces offline action—showing instead that internet connectivity acts as a catalyst for local democratic participation.